

Bias-Aware Job-Resume Matching: Auditing and Mitigating Demographic Bias in Automated Hiring Systems

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Abstract—Automated resume-screening systems are increasingly used to rank candidates before human review, but these systems may be sensitive to demographic cues that are unrelated to qualifications. In this project, we build a resume-job matching pipeline using Sentence-BERT and audit it with counterfactual resume variants. Each variant changes only one demographic signal, such as name, pronouns, or university, while keeping qualifications and experience fixed. We compute cosine similarity between resume and job embeddings and measure how much the score changes across counterfactual versions. Our results show that name changes produced the largest average score shift, followed by university and pronoun changes. To reduce this sensitivity, we implement blind screening by removing identity-related information before scoring. After blind screening, score differences are substantially reduced. We also include FLAN-T5-base, an open-source instruction-tuned language model, as a qualitative comparison for resume-job match decisions. Overall, this project shows that counterfactual auditing is useful for detecting demographic sensitivity in resume-matching systems and that simple mitigation can reduce this effect.

I. INTRODUCTION

Automated hiring tools are commonly used to filter and rank job applications before a human recruiter reviews them. These systems are useful because they can process large numbers of resumes quickly. However, they may also introduce or amplify unfairness if the model is sensitive to demographic cues that should not affect hiring decisions.

This project studies whether a resume-job matching model changes its similarity score when only one demographic-related signal is modified. The demographic signals examined in this project are applicant name, pronouns, and university. The main idea is simple: if two resumes have the same qualifications, skills, and experience, then the model should ideally assign them similar scores for the same job description.

We use Sentence-BERT as the main resume-job matching model because it converts text into semantic embeddings that can be compared using cosine similarity. We then use counterfactual resume variants to audit whether the model score changes when demographic cues are changed. In addition, we implement blind screening as a mitigation method and include FLAN-T5-base as an open-source LLM comparison, following

the project feedback to examine a current language model in addition to Sentence-BERT.

II. PROBLEM DESCRIPTION AND MOTIVATION

Automated resume-screening systems operate in a high-stakes decision-making setting. Even a small score difference can affect whether a candidate is shortlisted for further review. This makes fairness especially important. If a model assigns different scores to equally qualified applicants because of names, pronouns, or universities, then the system may unfairly affect hiring outcomes.

Prior work has shown that hiring can be influenced by demographic cues. Bertrand and Mullainathan [1] showed that resumes with white-sounding names received more callbacks than identical resumes with Black-sounding names. More recent work has also shown that language models can display demographic bias in resume-screening settings [4].

The motivation for this project is to build a practical machine learning pipeline that can audit this issue. Instead of only discussing bias conceptually, we build a resume-job matcher, generate controlled counterfactual resumes, calculate similarity scores, and measure whether those scores change when demographic signals are modified.

III. RESEARCH QUESTIONS

This project focuses on the following research questions:

- 1) Does a Sentence-BERT resume-job matching model assign different similarity scores to resumes that are identical except for one demographic signal?
- 2) Which demographic signal causes the largest score shift: name, pronouns, or university?
- 3) Does blind screening reduce the score differences across counterfactual resume versions?
- 4) How can an open-source instruction-tuned LLM provide a qualitative comparison to the Sentence-BERT matching pipeline?

These questions allow us to study both auditing and mitigation. The audit identifies whether demographic sensitivity exists,

while the mitigation step tests whether removing identity-related information reduces that sensitivity.

IV. RELATED WORK

This project connects two research areas: resume-job matching and algorithmic fairness in hiring. For resume-job matching, Qin et al. [2] proposed a neural approach for improving person-job fit. Reimers and Gurevych [3] introduced Sentence-BERT, which provides efficient sentence embeddings for semantic similarity tasks. We use Sentence-BERT as the main matching model because it can compare resumes and job descriptions based on meaning rather than exact keyword overlap.

On the fairness side, Bertrand and Mullainathan [1] showed that demographic cues in resumes can influence hiring outcomes. Wilson and Caliskan [4] studied gender, race, and intersectional bias in resume screening using language models. Our project follows a counterfactual auditing approach. We create controlled resume variants and measure whether the model score changes when only one demographic cue is modified.

V. DATASET

The dataset used in this project contains resumes and job descriptions across five domains:

- Software Engineering
- Finance
- Marketing
- Healthcare
- Education

The job dataset contains five job descriptions, one for each domain. The resume dataset contains 40 resume variants. Each base resume has an original version and counterfactual versions where one demographic signal is changed. The changed signals are name, pronouns, and university.

The important design choice is that the qualifications, skills, and experience remain fixed across the counterfactual versions. This means that if the score changes between an original resume and a modified version, the change can be connected to the demographic signal that was modified.

VI. METHODOLOGY

A. Sentence-BERT Matching

The main matching model uses Sentence-BERT with the all-MiniLM-L6-v2 model. Let r_i represent a resume and j_k represent a job description. Sentence-BERT converts each text into an embedding vector:

$$E_{r_i} = f_{\text{SBERT}}(r_i) \quad (1)$$

$$E_{j_k} = f_{\text{SBERT}}(j_k) \quad (2)$$

where E_{r_i} is the resume embedding, E_{j_k} is the job embedding, and f_{SBERT} is the Sentence-BERT encoder.

The resume-job matching score is computed using cosine similarity:

$$S(r_i, j_k) = \cos(E_{r_i}, E_{j_k}) = \frac{E_{r_i} \cdot E_{j_k}}{\|E_{r_i}\| \|E_{j_k}\|} \quad (3)$$

A higher cosine similarity score indicates that the resume is more semantically similar to the job description. Since the dataset contains 40 resume variants and 5 job descriptions, the total number of SBERT scores is:

$$40 \times 5 = 200 \quad (4)$$

B. Counterfactual Fairness Audit

For each original resume, we create counterfactual versions by changing only one demographic signal. Let r_i^{orig} be the original resume and r_i^{cf} be the counterfactual version. For the same job j_k , the original score is:

$$S_{\text{orig}} = S(r_i^{\text{orig}}, j_k) \quad (5)$$

The counterfactual score is:

$$S_{\text{cf}} = S(r_i^{\text{cf}}, j_k) \quad (6)$$

The signed score difference is:

$$\Delta_{i,k} = S_{\text{cf}} - S_{\text{orig}} \quad (7)$$

The absolute score difference is:

$$A_{i,k} = |\Delta_{i,k}| \quad (8)$$

The signed difference shows whether the counterfactual resume received a higher or lower score. The absolute difference shows the size of the score change, regardless of direction.

For each demographic signal g , the average absolute difference is:

$$\text{AvgAbsDiff}(g) = \frac{1}{N_g} \sum_{(i,k) \in G_g} |S(r_i^{\text{cf}}, j_k) - S(r_i^{\text{orig}}, j_k)| \quad (9)$$

where G_g is the set of comparisons where signal g was changed, and N_g is the number of such comparisons. A larger value means the model is more sensitive to that demographic signal.

C. Blind Screening Mitigation

Blind screening is used as a mitigation method. In this step, identity-related information is removed from the resume before scoring. Specifically, we remove the applicant’s name, pronouns, and university. Let $B(\cdot)$ represent the blind-screening transformation:

$$\tilde{r}_i = B(r_i) \quad (10)$$

where $\tilde{r}_i$ is the blind-screened resume.

The blind-screened score is computed as:

$$S_{blind}(\tilde{r}_i, j_k) = \cos(f_{SBERT}(\tilde{r}_i), f_{SBERT}(j_k)) \quad (11)$$

The blind-screened counterfactual score difference is:

$$\Delta_{i,k}^{blind} = S_{blind}(\tilde{r}_i^{cf}, j_k) - S_{blind}(\tilde{r}_i^{orig}, j_k) \quad (12)$$

The corresponding absolute difference is:

$$A_{i,k}^{blind} = |\Delta_{i,k}^{blind}| \quad (13)$$

To evaluate mitigation effectiveness, we compare the average absolute difference before and after blind screening:

$$R(g) = \frac{\text{AvgAbsDiff}_{before}(g) - \text{AvgAbsDiff}_{after}(g)}{\text{AvgAbsDiff}_{before}(g)} \quad (14)$$

where $R(g)$ is the reduction ratio for demographic signal g . A value closer to 1 indicates stronger reduction in demographic sensitivity.

D. LLM-Based Qualitative Comparison

To address the feedback about examining a current LLM, we include FLAN-T5-base, an open-source instruction-tuned language model. The LLM is used as a qualitative comparison, not as the main fairness metric.

For selected resume-job pairs, the LLM is prompted to classify the match as:

$$y_{LLM} \in \{\text{Strong Match}, \text{Partial Match}, \text{Weak Match}\} \quad (15)$$

This provides a human-readable judgment of resume-job fit. However, because generative model outputs can depend on prompt wording and generation settings, we keep Sentence-BERT as the main numerical model for fairness analysis.

VII. EXPERIMENTAL SETUP

All experiments were conducted in Google Colab using Python. The main libraries used were pandas, scikit-learn, Sentence-Transformers, and Hugging Face Transformers. The SBERT model used for scoring was all-MiniLM-L6-v2. The LLM used for qualitative comparison was FLAN-T5-base.

The experimental pipeline had four stages. First, the resume and job data were loaded and converted into text fields. Second, SBERT embeddings were generated for each resume and job description. Third, cosine similarity scores were calculated for all resume-job pairs. Fourth, the score differences between original and counterfactual resumes were computed.

For the blind-screening experiment, the same SBERT scoring process was repeated after removing name, pronouns, and university from the resume text. For the LLM experiment, selected original resume-job pairs were evaluated using FLAN-T5-base and classified into match levels.

VIII. EXPERIMENTAL RESULTS

A. SBERT Matching Results

The SBERT scoring pipeline successfully generated 200 resume-job similarity scores. Each score represents the semantic similarity between one resume variant and one job description. These scores were then used for counterfactual fairness analysis.

B. Counterfactual Fairness Results

Table I shows the average absolute score difference before blind screening.

Table I: Average absolute score difference before blind screening.

Changed Signal	Avg. Absolute Difference
Name	0.021594
Pronoun	0.006963
University	0.012347

The results show that changing the applicant’s name caused the largest average score shift. University changes had the second-largest effect, while pronoun changes had the smallest average effect. These differences show that the SBERT matching score is not completely insensitive to demographic-related text changes.

C. Blind Screening Results

After blind screening, the average absolute differences decreased. Table II shows the results after removing name, pronouns, and university from the resume text.

These results suggest that blind screening reduced the effect of demographic cues. Name and university differences were reduced to zero in this experiment, while pronoun-related differences also decreased.

Table II: Average absolute score difference after blind screening.

Changed Signal	Avg. Absolute Difference
Name	0.000000
Pronoun	0.003617
University	0.000000

D. Mitigation Reduction Analysis

Using the reduction formula, we can compare the model’s sensitivity before and after blind screening. For name changes, the average absolute difference decreased from 0.021594 to 0.000000. For university changes, it decreased from 0.012347 to 0.000000. For pronoun changes, it decreased from 0.006963 to 0.003617.

This means blind screening was especially effective for name and university signals in this dataset. The pronoun signal was also reduced, although not completely removed. This may happen because pronoun-related wording can still influence sentence structure or surrounding text in some resume variants.

E. LLM Comparison Results

The LLM comparison was used as a qualitative experiment. FLAN-T5-base classified selected original resume-job pairs into match levels such as strong match, partial match, or weak match. This helped us examine how an instruction-tuned language model responds to the resume-job matching task.

The LLM output is useful because it gives a human-readable decision. However, it is not used as the main fairness metric because LLM outputs can vary depending on the prompt. Instead, the LLM is included as an additional comparison to the SBERT-based scoring pipeline.

IX. DISCUSSION

The results show that counterfactual auditing is a useful method for testing whether a resume-matching model is sensitive to demographic signals. Since the resume variants are identical except for one changed signal, score differences provide evidence that the model is reacting to the changed demographic cue or to the text pattern created by that cue.

The name signal produced the largest average difference before mitigation. This is important because names are not qualification-related, but they may carry demographic associations. University changes also affected the score. This is more complex because university can be part of education history, but it can also act as a proxy for institutional prestige or socioeconomic background. Pronouns had the smallest effect, but the score difference was still measurable.

Blind screening reduced the counterfactual differences. This suggests that removing identity-related information can make the matching process less sensitive to demographic cues. However, blind screening is not a perfect solution. Removing university information may reduce bias, but it may also remove

context that some employers consider relevant. This creates a tradeoff between fairness and information preservation.

X. LIMITATIONS

This project has several limitations. First, the dataset is small, so the results should be interpreted as an audit-style experiment rather than a broad conclusion about real hiring systems. Second, the resume and job descriptions are simplified compared with real applicant tracking systems. Third, only one mitigation method, blind screening, was fully implemented. The original proposal also discussed data augmentation and adversarial debiasing, but those methods are left for future work.

The LLM component also has limitations. FLAN-T5-base was used only for qualitative match decisions, not for full numerical fairness analysis. Its outputs can be sensitive to prompt design, and it may not always give stable decisions across repeated runs or different wording.

XI. ETHICAL CONSIDERATIONS

This project focuses on an ethical problem in automated hiring. If a model assigns different scores to equally qualified applicants because of demographic cues, then it can unfairly influence who receives an interview opportunity. Even small score differences can matter when systems rank many applicants.

At the same time, this project does not claim to solve hiring bias. Names, pronouns, and universities are only proxies for demographic and social signals. Real identity and lived experience are more complex than these variables can represent. Blind screening can reduce some demographic sensitivity, but it does not remove deeper structural inequalities in hiring.

The project should therefore be understood as a fairness audit and mitigation study. The goal is to identify where score differences occur and test whether a simple mitigation step can reduce them.

XII. REPRODUCIBILITY

The code, data, and result files are organized in a public GitHub repository. The `data` folder contains the base resumes, resume variants, and job descriptions. The `notebooks` folder contains separate notebooks for SBERT scoring, fairness analysis, blind-screening mitigation, and the LLM comparison. The `results` folder contains the generated CSV files, including SBERT scores, fairness summaries, blind-screening results, and LLM match decisions.

The project is designed so that each notebook can be run independently in Google Colab. The SBERT notebook generates the original similarity scores. The fairness notebook computes counterfactual differences. The blind-screening notebook removes identity-related information and recalculates the scores. The LLM notebook uses FLAN-T5-base to produce qualitative match decisions.

XIII. FUTURE WORK

Future work can extend this project in several ways. First, the dataset can be expanded with more resumes, job descriptions, and industries. Second, data augmentation can be implemented by training or evaluating the model with more counterfactual resume pairs. Third, adversarial debiasing can be tested to reduce demographic information in the learned representation. Fourth, stronger LLMs can be evaluated more systematically for both match quality and bias. Finally, future work can compare fairness improvements with potential changes in matching accuracy.

XIV. CONCLUSION

This project built a bias-aware resume-job matching pipeline using Sentence-BERT, counterfactual fairness analysis, blind-screening mitigation, and an open-source LLM comparison. The results show that changing demographic signals can affect matching scores, especially for names and universities. Blind screening reduced these differences, suggesting that removing identity-related information can help reduce demographic sensitivity in this dataset. The LLM comparison provides an additional qualitative view of resume-job fit, while Sentence-BERT remains the main model for consistent numerical fairness analysis.

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